### ii) Common-Emitter (CE) configuration

In this configuration, the emitter is common to both the input and output circuits as shown in Figure 10.27. The base current $I_B$ is the input current and the collector current $I_C$ is the output current. The input signal is applied between emitter and base while the output is measured between collector and emitter.

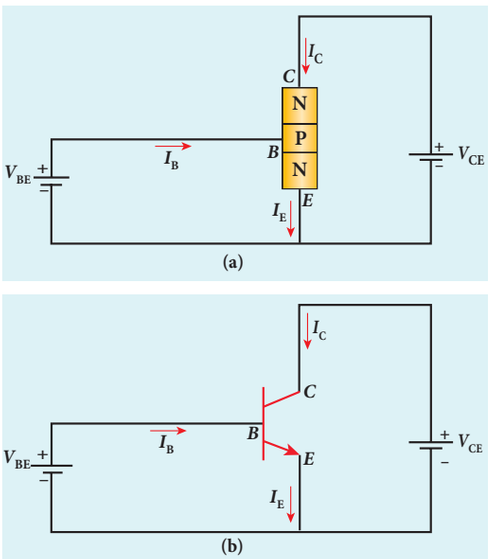



**Figure 10.27** NPN transistor in common emitter configuration (a) Schematic circuit diagram (b) Circuit symbol

### iii) Common-Collector (CC) configuration

Here, the collector is common to both the input and output circuits as shown in Figure 10.28. The base current $I_B$ is the input current and the emitter current $I_E$ is the output current. The input signal is applied between base and collector while the output is measured between emitter and collector.

**Note** As the output is taken from the emitter in common collector configuration, it is called an emitter follower.

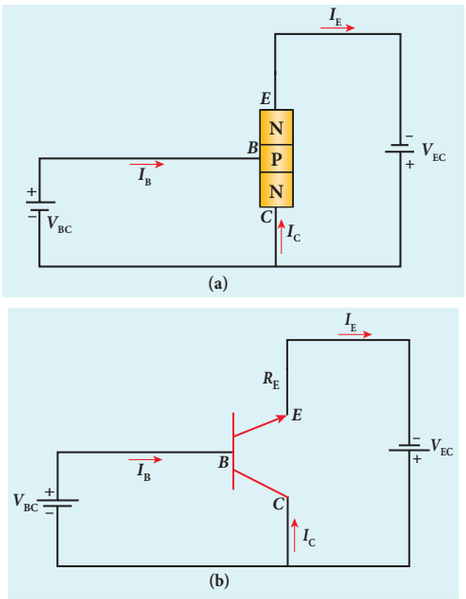



**Figure 10.28** NPN transistor in common collector configuration
(a) Schematic circuit diagram
(b) Circuit symbol

## 10.4.2 Transistor action in the common base mode

The operation of an NPN transistor in the common base mode is explained below. The current flow in a common base NPN transistor in the forward active mode is shown in Figure 10.29.

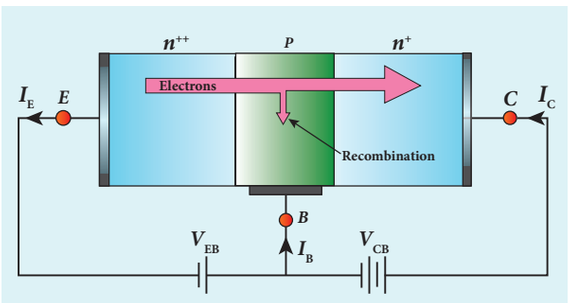



**Figure 10.29** Flow of current in a NPN transistor

Basically, a BJT can be considered as two *p-n* junction diodes connected back-to-back. In the forward active bias of the



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